LangChain & LangGraph Lab

Questions and Answers

Building LLM-powered applications and stateful AI agent workflows: core concepts, graph fundamentals and conditional routing evaluation

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LangChain & LangGraph

Core concepts, graph fundamentals and routing evaluation

Core Concepts — Roles of LangChain, LangGraph and an LLM

Q1. What is the role of LangChain?

LangChain is a framework for building LLM-powered applications by connecting language models with prompts, tools, and external data.

Q2. What is the role of LangGraph?

LangGraph is a framework for building stateful AI workflows using nodes, edges, and conditional routing.

Q3. What is the role of an LLM?

An LLM processes natural language inputs and generates responses. In an AI agent, it can also support reasoning, classification, and tool selection.

Expected result

```
{'message': 'Hello LangGraph!', 'response': 'Agent received: Hello LangGraph!'}
```

Exercise 2 — Graph Fundamentals

Q1. What is a node?

A node is a function that performs a task within the graph and can update the shared state.

Q2. What is an edge?

An edge defines the connection and execution flow between nodes.

Q3. What is `START`?

`START` is the entry point of the graph.

Q4. What is `END`?

`END` marks the completion of the graph execution.

Q5. Why is the state passed between nodes?

The state allows nodes to share information and preserve data throughout the workflow.

Routing Evaluation

Discussion prompt

> *Where did the router fail (if anywhere)? What would make the routing more robust (structured output, few-shot examples, enum validation)?*

Q1. Where did the router fail (if anywhere)?

- The router achieved **100% accuracy (6/6)** on the evaluation dataset.
- All mathematics, programming, and artificial intelligence questions were correctly classified.
- No routing errors were observed in this experiment.

Q2. What would make the routing more robust?

- **Structured output:** force the classifier to return a predefined category.
- **Few-shot prompting:** provide examples of correctly classified questions.
- **Enum validation:** reject unexpected categories.
- **Error handling:** provide a fallback when classification fails.
- **More testing:** evaluate the router using ambiguous and complex questions.

> **Conclusion**
> The conditional routing system performed successfully on all six test questions. However, a larger and more diverse evaluation dataset is necessary to assess its reliability.